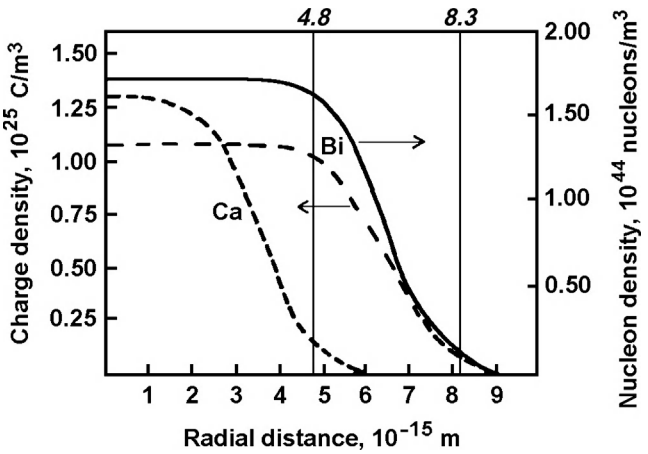


Figure 4.4 Experimentally measured charge and nuclear density values for ^{40}Ca and ^{209}Bi as a function of the nuclear radius.

charge and matter density near the surface of the irradiated nucleus can be studied. These experiments have led to the conclusion that nuclei do not possess a uniform charge or matter distribution out to a sharp boundary, but rather are fuzzy as indicated by the s-shaped curves in Figure 4.4. With an atomic number greater than 20 it has been found that a uniform charge and mass density exists over a short distance from the center of the nucleus, and this core is surrounded by a layer of decreasing density which seems to have a constant thickness of ~2.5 fm independent of mass number. In a bismuth nucleus, for example, the density remains relatively constant for approximately 5 fm then decreases steadily to one-tenth of that value in the next 2 fm (Fig. 4.4). This is not too surprising considering that as we get closer and closer to the edges of the nucleus not all the coordination sites of the nucleons are filled and thus the density will slowly decrease. It has also been found that not all nuclei are spherical, some being oblate and others prolate around the axis of rotation.

Despite the presence of this outer layer of decreasing density and the nonspherical symmetry, for most purposes it is adequate to assume a constant density nucleus with a sharp boundary. Therefore, use is made of the radius equation (4.14) in which the r_o value may be assumed to be 1.4 fm. Using this relationship, we can calculate the radius of ^{40}Ca to be $r = 1.4 \times 10^{-15} \times 40^{1/3} = 4.79$ fm, and for ^{209}Bi to be 8.31 fm. These values are indicated in Figure 4.4. For ^{80}Br a similar calculation yields 6.0 fm, while for ^{238}U the radius calculated is 8.7 fm. From these calculations we see that the radius does not change dramatically from relatively light nuclei to the heaviest.

4.6. SEMIEMPIRICAL MASS EQUATION

In preceding sections we have learned that the size as well as the total binding energy of nuclei are proportional to the mass number. These characteristics suggest an analogy

between the nucleus and a drop of liquid. In such a drop the molecules interact with their immediate neighbours but not with other molecules more distant. Similarly, a particular nucleon in a nucleus is attracted by nuclear forces only to its adjacent neighbours. Moreover, the volume of the liquid drop is composed of the sum of the volumes of the molecules or atoms present since these are nearly incompressible. Again, as we learned above, this is similar to the behavior of nucleons in a nucleus. Based on the analogy of a nucleus to a droplet of liquid, it has been possible to derive a semiempirical mass equation containing various terms which are related to a nuclear droplet.

Let us consider what we have learned about the characteristics of the nuclear droplet.

(a) First, recalling that mass and energy are equivalent, if the total energy of the nucleus is directly proportional to the total number of nucleons there should be a term in the mass equation related to the mass number.

(b) Secondly, in the discussion of the neutron/proton ratios we learned that the number of neutrons could not become too large since the discrepancy in the energy levels of the neutron and proton play a role in determining the stability of the nucleus. This implies that the binding energy is reduced by a term which allows for variation in the ratio of the number of protons and neutrons.

(c) Since the protons throughout the nucleus experience a mutual repulsion which affects the stability of nucleus, we should expect in the mass equation another negative term reflecting the repulsive forces of the protons.

(d) Still another term is required to take into account that the surface nucleons, which are not completely surrounded by other nucleons, would not be totally saturated in their attraction. In a droplet of liquid this lack of saturation of forces on the surface gives rise to the effect of surface tension. Consequently, the negative term in the mass equation reflecting this unsaturation effect should be similar to a surface tension expression.

(e) Finally, we have seen that nuclei with an even number of protons and neutrons are more stable than nuclei with an odd number of either type of nucleon and that the least stable nuclei are those for odd numbers of both neutrons and protons. This odd-even effect must also be included in a mass equation.

Taking into account these various factors, we can write a semiempirical mass equation. However, it is often more useful to write the analogous equation for the mass defect or binding energy of the nucleus, recalling (4.10). Such an equation, first derived by C. F. von Weizsäcker in 1935, would have the form:

$$E_{\mathrm{B}}(\mathrm{MeV}) = a_{\mathrm{v}}A - a_{\mathrm{a}}(N-Z)^2/A - a_{\mathrm{c}}Z^2/A^{1/3} - a_{\mathrm{s}}\,A^{2/3} \pm a_{\delta}/A^{3/4} \qquad (4.15)$$

The first term in this equation takes into account the proportionality of the energy to the total number of nucleons (the volume energy); the second term, the variations in neutron and proton ratios (the asymmetry energy); the third term, the Coulomb forces of repulsion for protons (the Coulomb energy); the fourth, the surface tension effect

(the surface energy). In the fifth term, which accounts for the odd-even effect, a positive sign is used for even proton-even neutron nuclei and a negative sign for odd proton-odd neutron nuclei. For nuclei of odd A (even-odd or odd-even) this term has the value of zero. Comparison of this equation with actual binding energies of nuclei yields a set of coefficients; e.g.

$$a_v = 15.5, \quad a_a = 23, \quad a_c = 0.72, \quad a_s = 16.8, \quad a_\delta = 34$$

With these coefficients the binding energy equations (4.5) and (4.10) give agreement within a few percent of the measured values for most nuclei of mass number greater than 40.

When the calculated binding energy is compared with the experimental binding energy, it is seen that for certain values of neutron and proton numbers, the disagreement is more serious. These numbers are related to the so-called "magic numbers", which we have indicated in Figure 4.1, whose recognition led to the development of the nuclear shell model described in a later chapter.

4.7. VALLEY OF β-STABILITY

If the semiempirical mass equation is written as a function of Z, remembering that $N = A - Z$, it reduces to a quadratic equation of the form

$$E_B = a\,Z^2 + b\,Z + c \pm d/A^{3/4} \tag{4.16}$$

where the terms a, b and c also contain A. This quadratic equation describes a parabola for constant values of A. Consequently, we would expect that for any family of isobars (i.e. constant A) the masses should fall upon a parabolic curve. Such a curve is shown in Figure 4.5. In returning to Figure 4.1, the isobar line with constant A but varying Z cuts

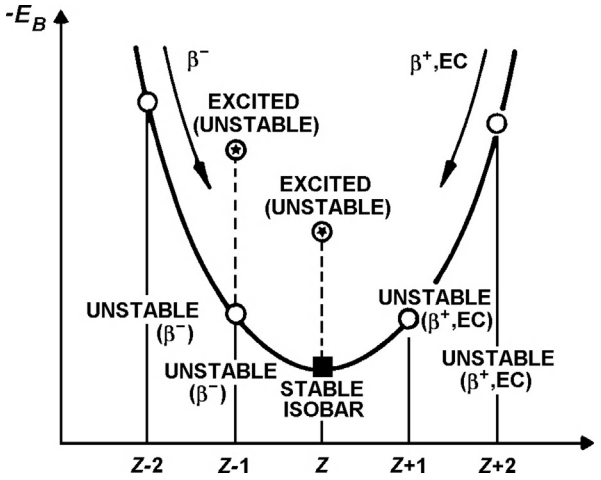


Figure 4.5 Isobar cut across the valley of stability showing schematically the position of different kinds of nuclei.